University of British Columbia

MECH 464/563 EECE 442/589 (Winter 2025): Introduction to Robotics

HW Assignment 3, Forward Kinematics

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Read Chapter 2, Forward Kinematics.

Exercise #1. (30 marks)

Consider the manipulator below. Assign joint variables in a manner consistent with the axes shown (positive angle by right hand rule), and coordinate systems $\{\underline{o}_i, \underline{C}_i\}$, to all links, using the Denavit-Hartenberg convention. Complete the table of Denavit Hartenberg parameters. Write, as a function of ${}^{i-1}T_i$, $i = 0, \ldots, 6$, the coordinate transformation relating the coordinates ${}^6x$ in the gripper frame with the coordinates ${}^0x$ in the base frame. You do not need to fill in the homogeneous matrices.

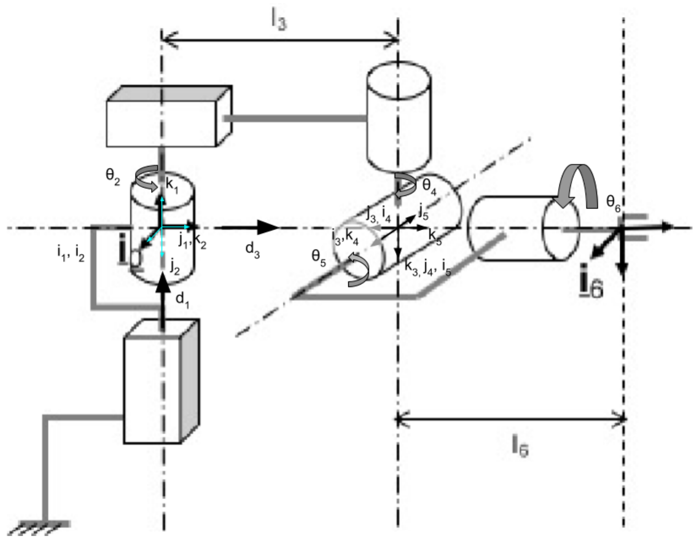

Figure 1: Cylindrical robot, with axes and co-ordinate systems added and labelled.

DH Parameters

DH Parameter	θ_i	d_i	a_i	α_i
Link 1	0	(d_1)	0	0
Link 2	(θ_2)	0	0	-90
Link 3	0	(d_3)	0	-90
Link 4	$(\theta_4) + 90$	0	0	90
Link 5	$(\theta_5) + 90$	0	0	-90
Link 6	-90	l_6	0	0

Transformation chain (Exercise #1)

$$ {}^{0}x = {}^{0}T_6\, {}^{6}x, \qquad {}^{0}T_6 = {}^{0}T_1\, {}^{1}T_2\, {}^{2}T_3\, {}^{3}T_4\, {}^{4}T_5\, {}^{5}T_6. $$

Exercise # 2: (30 marks for (a) and (b), 40 marks for (c))

The Puma 560 shown in Figure 2 has a reach of approximately 0.92m, and a payload capacity of 2.3kg. It was widely used for medium-to-lightweight assembly, welding, materials handling, packaging and inspection applications.

Using the schematic of Figure 3, do the following:

Figure 2: Photo of a 500 series Puma robot.

(a) Directly on the schematic, assign coordinate frames by means of the Denavit-Hartenberg convention. Assume C_0 and C_6 are as illustrated in the “home” position. Fill in Table 1 the values of the DH-parameters. For each joint, consider the positive rotation to be in the right-handed sense.

See Figure 3 for the co-ordinate frame placement.

Table 1: DH Parameters for the Puma 560

DH Parameter	θ_i (deg)	d_i (mm)	a_i (mm)	α_i (deg)
Link 1	(θ_1)	0	0	-90
Link 2	(θ_2)	0	430.0	180
Link 3	$(\theta_3)+90$	-149.1	20.3	90
Link 4	(θ_4)	435.0	0	90
Link 5	(θ_5)	0	0	-90
Link 6	(θ_6)	60.0	0	0

(b) Compose a chain of transformations that will give you the relationship between the base coordinate system $\{o_0, C_0\}$ and the end-effector coordinate system $\{o_6, C_6\}$. Use the same notation we have used for the Stanford manipulator (slides or p.7 Forward Kinematics notes).

Transformation chain

In the notation of page 7 in the notes:

$$\begin{bmatrix} \underline{C}_6 & \underset{\sim}{o}_6 \\ 0^T & 1 \end{bmatrix} = \begin{bmatrix} \underline{C}_6 & \underset{\sim}{o}_6 \\ 0^T & 1 \end{bmatrix} {}^0T_1(\theta_1)^1T_2(\theta_2)^2T_3(\theta_3)^3T_4(\theta_4)^4T_5(\theta_5)^5T_6(\theta_6)$$

$$= \begin{bmatrix} \underline{C}_6 & \underset{\sim}{o}_6 \\ 0^T & 1 \end{bmatrix} {}^0T_6(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6).$$

(c) Write a MATLAB m-file which allows the user to input the 6 joint angles (in degrees), and outputs both the resulting homogenous transformation matrix (relating the base and end effector coordinate systems). You will need to do direct kinematics from DH parameters for other robots, so you might as well include a function that takes an array of DH parameters and produces the gripper-to-base homogeneous transformation ${}^0T_n$. Demonstrate this (e.g., use diary) by computing the end-effector transformations for the following joint vector sets:

$$q_A = [0;0;0;0;0;0]^T, \quad q_B = [0;0;-90;0;0;180]^T, \quad q_C = [45;-45;45;0;-30;90]^T.$$

The computed end-effector transformations:

$$T_A = \begin{bmatrix} 0 & 0 & 1.0000 & 925.0000 \\ 0 & -1.0000 & 0 & 149.1000 \\ 1.0000 & 0 & 0 & 20.3000 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

$$T_B = \begin{bmatrix} -1.0000 & 0 & 0 & 450.3000 \\ 0 & 1.0000 & 0 & 149.1000 \\ 0 & 0 & -1.0000 & -495.0000 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}$$

$$T_C = \begin{bmatrix} 0.7071 & 0.6124 & -0.3536 & 74.0029 \\ -0.7071 & 0.6124 & -0.3536 & 284.8621 \\ 0 & 0.5000 & 0.8660 & 791.0174 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix}.$$

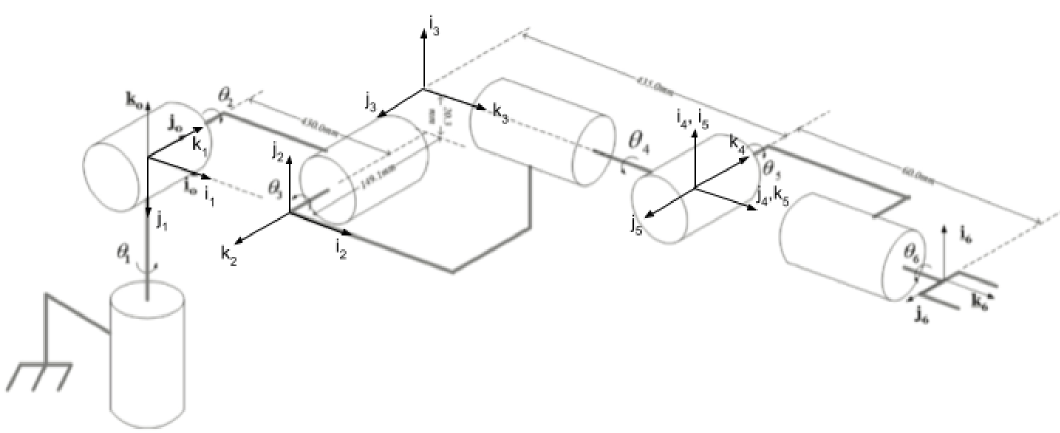

Figure 3: The Puma 560 robot schematic with co-ordinate frames (not to scale).

Matlab function:

```
function T = dh_fk(DH, q)
% Computes the transformation matrix given DH parameters (6*4 matrix) where
% angles are in degrees.
% parameters q = 1*n array with revolute joint angles (assumes only
% revolute joints), also in degrees.
    n = size(DH, 1); % Number of joints
    T = eye(4); % Initialize transformation matrix as identity
```

```
    for i = 1:n
        theta = (DH(i, 1) + q(i));
        d = DH(i, 2);
        a = DH(i, 3);
        alpha = DH(i, 4);
        % Formula for transformation given 4 DH parameters
        Tthet = eye(4);
        Tthet(1:3, 1:3) = rotz(theta);
        Td = eye(4);
        Td(3, 4) = d;
        Ta = eye(4);
        Ta(1, 4) = a;
        Talph = eye(4);
        Talph(1:3, 1:3) = rotx(alpha);
        T = T*Tthet * Td * Ta * Talph;
    end
end
```